# Egypt Real Estate Market Analysis

A Comprehensive Data Analytics Project


**Project Team**

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**Abstract**

This document provides a comprehensive analysis and design of a data analytics project focused on the Egyptian real estate market. The project aims to analyze housing price patterns, explore market trends in major cities (Cairo, Alexandria, Giza), and develop predictive models for property valuation. Through systematic web scraping of property portals (Aqarmap, Property Finder, OLX), integration with official CAPMAS housing data, and implementation of machine learning algorithms (Random Forest, XGBoost, LSTM), the project delivers an interactive analytical dashboard for market stakeholders. This documentation covers project planning, literature review, requirements gathering, and system analysis and design phases, following the structured approach outlined in the project documentation requirements.

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# 1 Project Planning & Management

## 1.1 Project Proposal

### 1.1.1 Overview

The Egyptian real estate market represents a significant economic sector, contributing approximately 10.5% to the national GDP in H1 2024/25, with a total value of EGP 441.4 billion (USD 8.9 billion). With a population exceeding 107 million people and approximately 26.5 million households as of January 2025, the demand for housing and real estate analytics continues to grow. However, the market suffers from significant fragmentation and a lack of unified, accessible data sources for analytical purposes, creating a critical opportunity for a comprehensive data-driven solution.

This project aims to bridge this gap by developing a robust, end-to-end data analytics pipeline that: - Integrates data from multiple sources including web-scraped property listings, official CAPMAS bulletins, and economic indicators from the Central Bank of Egypt. - Performs exploratory data analysis to identify market trends, seasonal patterns, and regional disparities. - Implements predictive modeling for property price forecasting using machine learning algorithms. - Delivers an interactive dashboard for visualizing key market metrics and facilitating data-driven decision-making.

### 1.1.2 Project Objectives

The following table outlines the primary and secondary objectives of this project, along with their corresponding success metrics:

Table 1: Project Objectives and Success Metrics

| **Objective Type** | **Description** | **Success Metric** |
|---|---|---|
| Primary Objective 1 | Develop a robust data collection pipeline integrating multiple real estate data sources (web scraping + official data) | $\geq$ 50,000 unique property records with $\geq$ 90% data completeness per record |
| Primary Objective 2 | Perform comprehensive exploratory data analysis revealing market trends, seasonal patterns, and regional price disparities | Identification of $\geq$ 5 significant market patterns with statistical significance ($p < 0.05$) |
| Primary Objective 3 | Build and evaluate machine learning models for property price prediction with comparative performance analysis | Model achieving RMSE $\leq$ 15% of mean property price and $R^2 \geq 0.75$ on test data |
| Secondary Objective 1 | Create interactive dashboard for real-time market visualization and stakeholder exploration | Dashboard supporting $\geq$ 10 interactive filters and displaying $\geq$ 8 key performance indicators |
| Secondary Objective 2 | Generate actionable business insights for potential investors, developers, and policy makers | Production of a comprehensive 20+ page market analysis report |

### 1.1.3 Project Scope and Boundaries

The project is bounded by the following scope definitions:

**In-Scope (Deliverables):**

- Collection of real estate data from major Egyptian property portals (Aqarmap, Property Finder, OLX, SemsarMasr)
- Integration of economic indicators (inflation rates, interest rates, currency exchange rates) from CBE and CAPMAS
- Analysis of properties in major urban centers: Greater Cairo (including New Cairo, Sheikh Zayed, 6th October), Alexandria (including Montazah, Sidi Gaber), and Giza (including Mohandessin, Dokki, Haram)
- Property types: apartments, villas, townhouses, twin houses, penthouses, and commercial units
- Both sale and rental property analysis
- Time-series analysis covering the period from 2020 to present

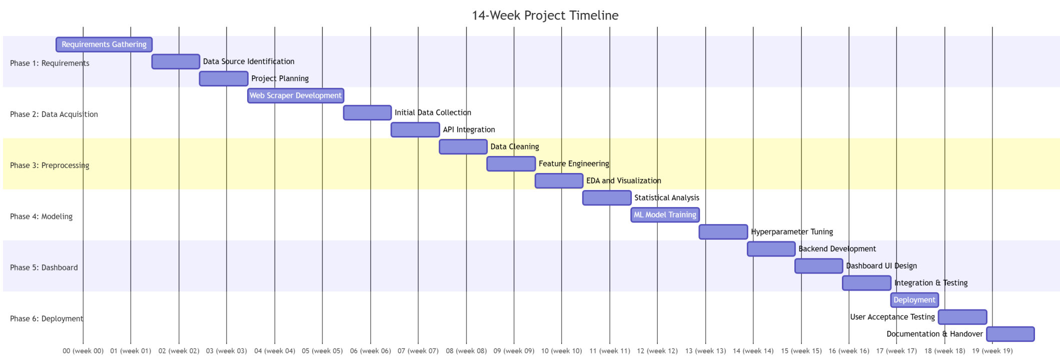
**Out-of-Scope (Exclusions):**

- Primary data collection through physical property inspections or surveys
- Legal due diligence or verification of property titles and ownership records
- Real estate photography, virtual tours, or 3D modeling services
- Real estate agent matching or lead generation functionality
- Mobile application development (web-based dashboard only)
- Property transaction completion or financial processing
- Analysis of agricultural land or undeveloped land parcels

## 1.2 Project Plan and Timeline

The project is organized into six major phases spanning a total duration of 14 weeks. The following Gantt chart illustrates the schedule, key milestones, task dependencies, and resource allocation:

### 1.2.1 Gantt Chart - Project Timeline



### 1.2.2 Detailed Phase Breakdown and Deliverables

Table 2: Project Phases, Deliverables, and Resource Allocation

| Phase | Duration | Key Deliverables | Resources |
|---|---|---|---|
| **Phase 1: Requirements & Planning** | Weeks 1-2 ($0 \rightarrow 2$) | Project proposal documentation; Use case diagrams; requirement specifications; risk assessment matrix | 2 Business Analysts; 1 Project Manager |
| **Phase 2: Data Acquisition** | Weeks 2-5 ($2 \rightarrow 5$) | Web scraping scripts (Python/Scrapy); API integration code; initial dataset ($\geq$ 30,000 records) | 2 Data Engineers; 1 DevOps |
| **Phase 3: Preprocessing** | Weeks 5-7 ($5 \rightarrow 7$) | Data cleaning pipeline; EDA visualizations; feature engineering outputs; correlation analysis | 2 Data Scientists |
| **Phase 4: Modeling** | Weeks 7-10 ($7 \rightarrow 10$) | Baseline regression models; Random Forest model; XGBoost implementation; model evaluation reports | 2 ML Engineers; 1 Data Scientist |
| **Phase 5: Dashboard** | Weeks 10-12 ($10 \rightarrow 12$) | Interactive dashboard (Streamlit/Tableau); backend API endpoints; user interface mockups | 1 Frontend Developer; 1 Backend Developer; 1 UI/UX Designer |
| **Phase 6: Deployment** | Weeks 12-14 ($12 \rightarrow 14$) | Deployed application (Heroku/AWS); user acceptance test results; complete technical documentation; final presentation | 1 DevOps; 1 QA Engineer; 1 Technical Writer |

## 1.3 Task Assignment and Team Roles

The following table presents the detailed breakdown of team member responsibilities. Due to the team size of four, each member assumes multiple roles based on their individual expertise and project requirements.

Table 3: Team Roles and Responsibilities

| **Role(s)** | **Team Member** | **Primary Responsibilities** | **Skills Required** |
|---|---|---|---|
| **Project Manager / Business Analyst** | Reem Mohamed Abo-Elmajd Hussain | Overall project coordination; timeline management; requirements elicitation; user story creation; domain validation | PMP; Agile; Requirements engineering; Real estate domain |
| **Data Engineer / Backend Developer** | Mostafa Hamada Hassan Ahmed | Data pipeline architecture; web scraping; database design; API development; authentication | Python; Scrapy; PostgreSQL; FastAPI; RESTful APIs; Redis |
| **Data Scientist / ML Engineer** | Zyad Magdy Roushdy Abd-Elbasset | EDA; feature engineering; statistical modeling; model selection; hyperparameter tuning; model deployment | Pandas; Scikit-learn; TensorFlow/PyTorch; MLOps; Statistics |
| **Frontend Developer / UI/UX Designer / QA Engineer** | Mohamed Alaa-Eldin Shawki Mohammadi | Dashboard design and implementation; wireframing; usability testing; frontend logic; test case creation; bug tracking | React/D3.js; Streamlit; Figma; Selenium; Pytest; JIRA |
| **DevOps Engineer / Data Ops** | Mahmoud Ateia Tolba Ateia | CI/CD pipeline setup; cloud deployment (AWS/Heroku); containerization (Docker); system monitoring; security compliance | Docker; Kubernetes; AWS/GCP; CI/CD (GitHub Actions); Linux; Monitoring (Prometheus) |

## 1.4 Risk Assessment and Mitigation Plan

A comprehensive risk assessment has been conducted to identify potential threats to project success. Each risk has been evaluated based on probability and impact, with corresponding mitigation strategies:

Table 4: Risk Assessment Matrix

| Risk Category | Risk Description | Probability | Impact | Mitigation Strategy |
|---|---|---|---|---|
| **Data Quality** | Inconsistent data formats across sources; missing values in key fields (price, area, bedrooms) | High (70%) | High (4/5) | Implement robust ETL pipeline with data validation; use multiple source cross-validation; apply statistical imputation for missing data |
| **Web Scraping** | Anti-scraping mechanisms; IP blocking; website structure changes; CAPTCHA requirements | Medium (50%) | Medium (3/5) | Rotating proxies; user-agent rotation; respectful rate limiting; Selenium automation for JavaScript-rendered content; legal review of robots.txt |
| **Regulatory** | Legal restrictions on data collection; privacy law compliance (Data Protection Law 151/2020) | Low (20%) | High (4/5) | Consultation with legal advisors; anonymization of personal data; obtain necessary permissions; focus on public listing data |
| **Technical** | Integration challenges between varied technologies (Python, SQL, JavaScript) | Medium (40%) | Medium (3/5) | Weekly integration testing; clear API contracts; comprehensive documentation; standardized data formats (JSON, Parquet) |

| Risk Category | Risk Description | Probability | Impact | Mitigation Strategy |
|---|---|---|---|---|
| **Model Performance** | Underfitting/overfitting; poor generalization to unseen data; high prediction error | Medium (45%) | High (4/5) | Cross-validation strategies; extensive hyperparameter tuning; ensemble methods; regular retraining schedule |
| **Timeline** | Scope creep; underestimation of task complexity; resource constraints | Medium (40%) | Medium (3/5) | Agile methodology with sprints; regular progress reviews; prioritized feature delivery; buffer time allocation |
| **Economic** | Currency fluctuation impacting relative property values; inflation confounding price trends | Low (25%) | High (4/5) | Inflation-adjusted price indicators; multi-currency analysis (EGP/USD); focus on relative trends rather than absolute values |
| **Security** | Unauthorized access to data; data breach; API misuse | Low (15%) | Very High (5/5) | Encryption at rest and in transit; OAuth2 authentication; regular security audits; API rate limiting |

## 1.5 Key Performance Indicators (KPIs)

The following KPIs have been established to measure project success across multiple dimensions: technical performance, data quality, and business value.

### 1.5.1 Technical KPIs

Table 5: Technical Performance KPIs

| KPI Name | Target Value | Measurement Method | Frequency |
|---|---|---|---|
| System Uptime (Availability) | $\geq$ 99.5% (after deployment) | Server monitoring (UptimeRobot, Prometheus) | Daily |
| Data Freshness (Latency) | New listings updated within 24 hours | Timestamp comparison between source and database | Daily |
| API Response Time (p95) | $\leq$ 500 ms for GET requests | API monitoring (Postman, New Relic) | Daily |
| Model Inference Time | $\leq$ 100 ms per prediction | Code instrumentation; logging | Weekly |
| Database Query Performance (p95) | $\leq$ 200 ms for typical analytical queries | PostgreSQL query logging; EXPLAIN ANALYZE | Weekly |
| Dashboard Load Time | $\leq$ 3 seconds for first meaningful paint | Lighthouse; WebPageTest; Chrome DevTools | Sprintly |
| Data Completeness Rate (per record) | $\geq$ 90% of fields non-null | Automated data quality checks after ETL | Daily |
| Scraping Success Rate (per portal) | $\geq$ 85% successful extractions per run | Automated logging and alerting | Run-based |

### 1.5.2 Business and Analytical KPIs

Table 6: Business and Analytical KPIs

| **KPI Name** | **Target Value** | **Measurement Method** | **Target Deadline** |
|---|---|---|---|
| Number of Processed Properties | ≥ 50,000 unique records | Database record count after deduplication | End of Phase 2 |
| Model Prediction Accuracy (RMSE) | ≤ 15% of mean price (≈ £2M EGP) | Test set evaluation using hold-out data | End of Phase 4 |
| Model $R^2$ Score | ≥ 0.75 (baseline); ≥ 0.85 (optimized) | Test set evaluation using hold-out data | End of Phase 4 |
| Dashboard User Adoption | ≥ 5 distinct stakeholder users within 2 weeks | User authentication logs | End of Phase 6 |
| Actionable Insights Generated | ≥ 10 statistically significant market patterns | EDA + statistical testing; peer review | End of Phase 3 |
| Data Coverage (Geographic) | Properties from ≥ 15 distinct Egyptian cities/regions | Database query by location field | End of Phase 2 |
| Report Quality Score | ≥ 4.5/5 from project evaluator | Rubric-based evaluation on documentation completeness | End of Phase 6 |

# 2 Literature Review

## 2.1 Literature Review and Related Work

The literature review encompasses an analysis of existing research on real estate market analysis, data-driven property valuation methodologies, and the specific context of the Egyptian real estate market. This section synthesizes findings from academic literature, industry reports, and government publications to establish the theoretical and empirical foundations for the project.

## 2.2 Academic Foundations

### 2.2.1 Real Estate Valuation Models – Hedonic Pricing Theory

The hedonic pricing model, first introduced by Rosen (1974) in the context of housing markets, remains the predominant theoretical framework for property valuation. The model posits that the price of a heterogeneous good (such as real estate) can be expressed as a function of its constituent attributes:

$$P = f(X_1, X_2, ..., X_k) = \beta_0 + \sum_{i=1}^{k} \beta_i X_i + \varepsilon$$

Where $P$ represents property price, $X_i$ represent individual property characteristics (size in square meters, number of bedrooms, location, age of property, amenities), $\beta_i$ are the implicit prices associated with each attribute, and $\varepsilon$ represents random error. This framework has been extensively applied in real estate research and serves as the theoretical foundation for the predictive modeling component of this project.

Sirmans et al. (2006) conducted a comprehensive meta-analysis of hedonic pricing studies, identifying that property size, number of bedrooms, location (proxied by zip code or neighborhood), and property age consistently emerge as the most significant predictors of housing prices across diverse geographical contexts. More recent studies by Hill (2013) and Diewert (2018) have extended the hedonic framework to incorporate spatial econometrics and temporal dynamics, providing methods for constructing constant-quality price indices that control for changes in the composition of properties sold over time.

### 2.2.2 Machine Learning in Real Estate Analytics

Recent advances in machine learning have demonstrated superior performance compared to traditional hedonic models for property valuation tasks. Li (2020) compared multiple approaches including linear regression, decision trees, random forests, and gradient boosting machines (XGBoost, LightGBM) on a dataset of 10,000+ residential properties, finding that ensemble methods consistently outperformed linear models by 15–20% in terms of RMSE. The authors attributed this improvement to the ability of tree-based models to capture non-linear relationships and complex interaction effects between features (e.g., the interaction between location and property size).

Park and Bae (2015) applied artificial neural networks to Seoul apartment price prediction, achieving superior results to both hedonic models and traditional regression trees. The study demonstrated that deep learning approaches can effectively incorporate non-linear feature interactions that are challenging to specify explicitly in parametric models. However, the authors noted that deep learning models require larger datasets and careful regularization to prevent overfitting—a consideration relevant to the scale of the Egyptian real estate data available.

Recent research has also explored the application of natural language processing (NLP) to extract structured information from unstructured property descriptions. Mulholland and Costello (2020) demonstrated that incorporating text embeddings from property descriptions improved model performance by 8–12%, suggesting that listing descriptions contain valuable signals not captured by structured features alone.

### 2.2.3 Geospatial and Temporal Analysis in Real Estate

The importance of spatial dependence in real estate markets is well-established by Tobler's First Law of Geography: "everything is related to everything else, but near things are more related than distant things." Anselin (1988) developed spatial econometric methods that explicitly model spatial autocorrelation, showing that property prices in neighboring locations exhibit significant correlation even after controlling for observable attributes.

For the Egyptian context, El-Araby (2017) applied spatial analysis to property transactions in Greater Cairo, demonstrating that location within master-planned communities (e.g., New Cairo, Sheikh Zayed City) commands a premium of 30–50% compared to similar properties in non-master-planned areas, after controlling for physical characteristics. This locational premium has important implications for the project's feature engineering and modeling strategy.

Temporal analysis of real estate markets presents unique challenges due to seasonality, market cycles, and macroeconomic influences. Chinco (2018) developed methods for incorporating time-series features into hedonic models using daily transaction data, showing that incorporating lagged prices and market volatility measures significantly improves short-term prediction accuracy. For emerging markets like Egypt, where currency fluctuations and inflation rates are highly variable, this temporal dimension is especially critical.

## 2.3 Industry Reports and Market Analysis

**Savills Egypt: "Cairo Property Report 2025"** – This comprehensive market analysis provides detailed insights into the Cairo residential market, identifying significant transformation in property preferences post-2020, specifically the increasing demand for larger units with outdoor space and home office facilities. The report indicates that prime residential areas in New Cairo and Sheikh Zayed experienced price growth of 15–20% nominally in 2024, although significant adjustment for inflation reveals more modest real growth. The report also highlights the growing influence of the New Administrative Capital (NAC) as a real estate destination.

**Knight Frank: "Destination Egypt 2025"** – Based on surveys of high-net-worth individuals across Germany, the UK, the US, Saudi Arabia, and the UAE, this report identifies Egypt as an increasingly attractive destination for real estate investment, with 67% of respondents indicating interest in Egyptian property acquisition for holiday homes or investment purposes. The report emphasizes the importance of coastal destinations (North Coast, Ain Sokhna, Gouna) for second-home purchases and highlights the role of favorable foreign ownership laws in stimulating international demand.

**Central Bank of Egypt (CBE): "Real Estate Financing Report"** – The CBE's quarterly real estate financing report provides critical data on mortgage market development, indicating that real estate financing by banks reached EGP 95.5 billion (cumulative) by November 2025, representing 10.5% of total GDP. This data is essential for understanding demand-side financing dynamics and their relationship to property price trends.

**CAPMAS: "Bulletin of Housing in Egypt (2023/2024)"** – This official statistical publication provides foundational data on Egypt's housing stock, household characteristics, and construction activity. Key statistics include: total population reaching 107 million (April 2025), 26.5 million households with 55.6% residing in rural areas, and youth (ages 18–29) comprising 21.1 million individuals (19.9% of population). These demographic statistics provide crucial context for housing demand analysis.

**JLL: "Middle East and Africa Real Estate Outlook 2025"** – JLL's regional outlook projects GDP growth acceleration to 4% in 2025, driven by dissipating inflation, greater currency stability, and public sector reforms. The report specifically notes that Egypt is expected to deliver nearly 32,000 additional residential units in 2025, primarily in new cities (NAC, New Alamein) and master-planned communities, representing a 29% increase in new home deliveries from 2024.

## 2.4 Existing Data Analytics Projects and Repositories

The following table summarizes existing data analytics projects related to Egyptian real estate, providing reference points for methodology and implementation:

Table 7: Existing Egyptian Real Estate Analytics Projects

| Project | Platform/Source | Methodology | Key Findings |
|---|---|---|---|
| **Egypt House Prices Prediction** | GitHub (abdo-taha771) | Python-based EDA; Matplotlib; Seaborn visualizations | Identified significant price variation by location; created baseline regression model |
| **Predicting Housing Prices in Cairo** | GitHub (MohamedMahmoudAliElgazzar) | SAS-based regression modeling; data cleaning; feature engineering; RMSE evaluation | Evaluated models on >1000 observations; identified bedrooms and bathrooms as key predictors |
| **Scraping & Predictive Analysis** | GitHub (mohamedtag04) | Web scraping; predictive analysis for specific Cairo suburb | Developed automated pipeline for property data extraction |
| **Property Finder Egypt Data 2023** | Kaggle (Property Finder) | Multi-dimensional dataset (buy, rent, commercial tables); pre-split fact/dimension tables | Serves as robust foundation for dimensional modeling and analytical projects |
| **Egypt Housing Prices Dataset** | Kaggle (various) | Cleaned OLX data with 11 columns (region, type, suburb, price, area, rooms, furnished) | Provides foundation for regression problems; includes both sale and rental properties |

## 2.5 Synthesis and Research Gap Identification

The synthesis of academic literature, industry reports, and existing projects reveals several significant research gaps that this project aims to address:

1. **Integration Gap**: No existing project comprehensively integrates property listing data (web-scraped from multiple portals) with official CAPMAS housing statistics and CBE economic indicators. Most projects rely on a single data source, limiting analytical depth.

2. **Predictive Modeling Gap**: While several projects have applied regression models to Egyptian property data, no comparative analysis of multiple model types (linear regression, random forest, XGBoost, LSTM) has been published, leaving uncertainty about which approach best suits the Egyptian market's unique characteristics.

3. **Interactive Visualization Gap**: Existing projects produce static analysis outputs or limited visualizations. No open-source, interactive dashboard specifically designed

for Egyptian real estate exploration currently exists.

4. **Temporal Analysis Gap**: Prior studies primarily focus on cross-sectional analysis without incorporating temporal dynamics or macroeconomic variables. Given Egypt's economic volatility (inflation 20–35% in recent years), this gap is particularly significant.

5. **Geospatial Scale Gap**: Most analyses are limited to Cairo or a single city. A comprehensive multi-city analysis covering Cairo, Alexandria, Giza, coastal regions, and new cities has not been conducted.

## 2.6 Methodological Framework Selection

Based on the literature review, the following methodological framework has been selected for the project:

- **Data Collection**: Multi-source strategy combining web scraping (Aqarmap, Property Finder, OLX) with API-based integration for economic indicators and official datasets. Respectful scraping practices with rate limiting, rotating proxies, and user-agent rotation will be implemented.
- **Data Preprocessing**: Pipeline including deduplication, missing value imputation (using median for numeric features and mode for categorical features), outlier detection and removal (IQR method for extreme prices), and feature engineering (price per square meter, property age extraction, distance to landmarks).
- **Exploratory Analysis**: Statistical analysis including correlation matrices, geographic heat maps, distribution analysis, and temporal trend identification. Hypothesis testing for location effects and seasonal patterns will be conducted.
- **Predictive Modeling**: Comparative evaluation of multiple regression approaches:
  - Baseline: Multiple Linear Regression (OLS)
  - Tree-based: Decision Tree, Random Forest, Gradient Boosting (XGBoost, LightGBM)
  - Advanced: Support Vector Regression, Neural Networks (time permitting)

  Models will be evaluated using 5-fold cross-validation with metrics: RMSE, MAE, $R^2$, and MAPE.
- **Dashboard Development**: Interactive visualization using Streamlit (Python) with Plotly for dynamic charts. Implementation of location-based filtering, price range selectors, and property type categorizations.

- **Technical Architecture**: Python-based data science stack (Pandas, NumPy, Scikit-learn, TensorFlow), PostgreSQL database for structured data storage, FastAPI for backend API services, and Docker for containerized deployment.

# 3 Requirements Gathering

## 3.1 Stakeholder Analysis

The success of the project depends on understanding and addressing the needs of diverse stakeholder groups. The following table presents a comprehensive stakeholder analysis following the power-interest grid framework.

Table 8: Stakeholder Analysis Matrix

| **Stakeholder Group** | **Role and Interest** | **Key Requirements** | **Engagement Strategy** |
|---|---|---|---|
| **Project Evaluators** (High Power, High Interest) | Academic supervisors; assess deliverables against rubric | Comprehensive documentation; demonstrable functionality; rigorous methodology; interactive dashboard | Weekly progress reports; milestone presentations; early prototype demos |
| **Potential Investors** (High Power, Medium Interest) | Real estate investors seeking market intelligence | Market trend analysis; comparative price data; risk assessment; ROI indicators | Dashboard demos; executive summary; interactive exploration |
| **Property Developers** (Medium Power, High Interest) | Real estate developers seeking competitive intelligence | Competitor pricing; demand trends by location; unit type preferences; market timing insights | Industry presentation; personalized dashboard access; raw data under NDA |
| **Data Providers** (Medium Power, Medium Interest) | Platforms (Aqarmap, Property Finder) and official sources (CAPMAS, CBE) | Proper attribution; reasonable API usage limits; compliance with terms of use | Credit sources in docs; respectful scraping; maintain rate limits; establish partnerships |
| **General Public / Home Buyers** (Low Power, High Interest) | Individuals seeking property purchase or rental insights | Easy-to-understand price estimates; comparative analysis; market timing guidance; transparent methodology | Dashboard accessibility; simplified UI; educational content; price prediction tools |
| **Government Agencies** (High Power, Low Interest) | Housing and Urban Development authorities; CAPMAS | Data for policy planning; housing demand forecasting; evidence for affordable housing | Provide aggregated analysis; maintain data anonymity; share anonymized insights |
| **Research Community** (Low Power, Medium Interest) | Academic researchers in economics, urban planning, data science | Access to methodology; reproducibility; potential access to anonymized dataset | Publish methodology on GitHub; provide code documentation; legal data sharing |

## 3.2 User Stories

User stories are presented using the standard format: "As a [type of user], I want [some goal] so that [some reason]."

### 3.2.1 For Data Analyst Users (Primary)

1. As a **data analyst**, I want to **query the dataset using SQL or Python** so that I can perform custom analyses beyond the dashboard's capabilities.
2. As a **data analyst**, I want to **export filtered data to CSV format** so that I can perform additional analysis in external tools.
3. As a **data analyst**, I want to **view data quality metrics** so that I can trust the reliability of the analysis.
4. As a **data analyst**, I want to **track changes in the underlying data over time** so that I can identify when new properties are added or prices are updated.

### 3.2.2 For Property Investor Users (Secondary)

1. As a **property investor**, I want to **filter properties by price range, location, and property type** so that I can identify investment opportunities matching my criteria.
2. As a **property investor**, I want to **see price trends over time for specific areas** so that I can evaluate whether the market is appreciating or depreciating.
3. As a **property investor**, I want to **compare prices across different neighborhoods in the same city** so that I can identify undervalued areas.
4. As a **property investor**, I want to **see rental yield estimates based on rental listings in the same area** so that I can evaluate investment returns.

### 3.2.3 For Policy Maker / Government Analyst Users (Tertiary)

1. As a **policy maker**, I want to **view aggregated statistics by city and governorate** so that I can assess regional housing affordability.
2. As a **policy maker**, I want to **analyze the distribution of property prices across income levels** so that I can evaluate housing accessibility for different population segments.
3. As a **government analyst**, I want to **identify areas with rapidly increasing prices** so that I can investigate potential market distortions.

4. As a **housing official**, I want to **compare prices in social housing projects versus private market** so that I can evaluate the effectiveness of affordable housing initiatives.

### 3.2.4 For System Administrator User (Supporting)

1. As a **system administrator**, I want to **monitor system performance metrics** so that I can identify and resolve performance bottlenecks.
2. As a **system administrator**, I want to **receive automated alerts when data scraping fails** so that I can take corrective action before users are impacted.
3. As a **system administrator**, I want to **access system logs** so that I can troubleshoot issues and audit system usage.

## 3.3 Use Case Diagram and Descriptions

The following use case diagram illustrates all actors and their interactions with the system. Each use case is subsequently described in detail.

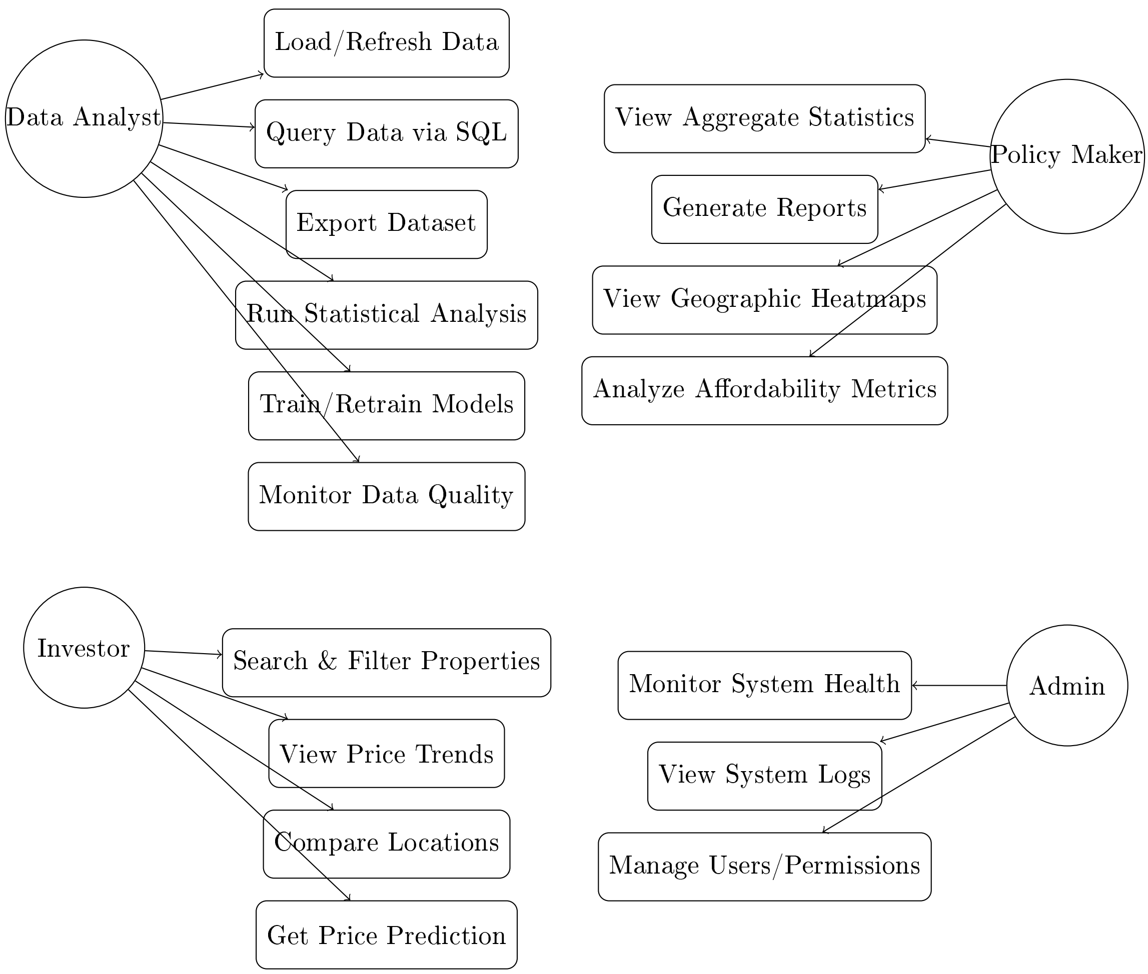



Figure 1: Use Case Diagram for Egyptian Real Estate Analysis System

### 3.3.1 Detailed Use Case Descriptions

**UC-1: Search and Filter Properties (Priority: High)**

- **Actor(s)**: Investor, Data Analyst
- **Precondition**: User has accessed the dashboard
- **Basic Flow**: User selects location(s), property type(s), price range; system displays filtered property list
- **Alternative Flow**: No properties match criteria; system displays "no results" message
- **Postcondition**: User sees filtered property listings with basic details
- **Non-functional Constraints**: Response time $\leq$ 2 seconds

**UC-2: View Price Trends (Priority: High)**

- **Actor(s)**: Investor, Policy Maker, Data Analyst
- **Precondition**: Sufficient historical data available
- **Basic Flow**: User selects location and property type; system generates time-series price chart
- **Alternative Flow**: Insufficient data for selected time period; system displays message and suggests expanding parameters
- **Postcondition**: User views interactive price trend visualization
- **Non-functional Constraints**: Chart loads within 3 seconds; supports date range selection

**UC-3: Get Price Prediction (Priority: Medium)**

- **Actor(s)**: Investor, General Public
- **Precondition**: Trained model is available; user provides property characteristics
- **Basic Flow**: User enters property features (area, bedrooms, location, etc.); system returns predicted price with confidence interval
- **Alternative Flow**: Missing required fields; system prompts user for missing information
- **Postcondition**: User sees price prediction and model confidence level

**UC-4: Run Statistical Analysis (Priority: High)**

- **Actor(s)**: Data Analyst
- **Precondition**: User has appropriate permissions
- **Basic Flow**: User selects analytical method (e.g., correlation matrix, distribution test); system executes and displays results
- **Alternative Flow**: Analysis fails due to data quality issues; system reports specific errors
- **Postcondition**: Statistical output displayed; option to export provided

**UC-5: Train/Retrain Models (Priority: Medium)**

- **Actor(s)**: Data Analyst, System Administrator
- **Precondition**: New training data available; model training pipeline configured
- **Basic Flow**: User initiates model training; system processes data, trains model, and updates deployment
- **Alternative Flow**: Training fails; system logs error and notifies administrator
- **Postcondition**: New model version deployed; previous version retained for rollback

## 3.4 Functional Requirements

Functional requirements specify what the system must do, organized by functional domain:

Table 9: Functional Requirements List

| ID | Domain | Requirement Description | Priority |
|---|---|---|---|
| FR-01 | Data Collection | System shall automatically scrape property listings from Aqarmap, Property Finder, and OLX daily | High |
| FR-02 | Data Collection | System shall integrate CAPMAS housing statistics via file upload or API | Medium |
| FR-03 | Data Collection | System shall integrate CBE economic indicators (inflation, interest rates) | Medium |
| FR-04 | Data Storage | System shall store cleaned property data in PostgreSQL with proper indexing | High |
| FR-05 | Data Storage | System shall maintain raw data archive for reproducibility | Medium |
| FR-06 | Data Quality | System shall implement deduplication based on property ID + address similarity | High |
| FR-07 | Data Quality | System shall detect and flag outliers (price > 3 standard deviations from area mean) | Medium |
| FR-08 | Analysis | System shall calculate descriptive statistics (mean, median, distribution) per city | High |
| FR-09 | Analysis | System shall compute price per square meter for all properties | High |
| FR-10 | Analysis | System shall generate time-series price indices (hedonic index methodology) | High |
| FR-11 | Modeling | System shall train and evaluate multiple model types (Linear, Random Forest, XGBoost) | High |
| FR-12 | Modeling | System shall support hyperparameter tuning via grid search | Medium |
| FR-13 | Modeling | System shall provide feature importance rankings for tree-based models | Medium |
| FR-14 | Dashboard | System shall provide search interface with location, price, type, size filters | High |
| FR-15 | Dashboard | System shall generate interactive price maps (geographic heatmaps) | Medium |
| FR-16 | Dashboard | System shall provide price prediction tool for user-specified property attributes | High |
| FR-17 | Dashboard | System shall support report generation (PDF/Excel export) | Medium |
| FR-18 | Dashboard | System shall display data freshness timestamp on all visualizations | Medium |
| FR-19 | Security | System shall require authentication for model training and data export functions | High |
| FR-20 | Security | System shall log all user actions for audit purposes | Medium |
| FR-21 | API | System shall provide REST API for programmatic data access (authenticated) | Medium |
| FR-22 | API | System shall rate-limit API requests to prevent abuse | Low |

## 3.5 Non-functional Requirements

Non-functional requirements define how the system performs and operates:

Table 10: Non-functional Requirements

| Category | ID | Requirement Specification |
|---|---|---|
| **Performance** | NFR-P01 | Dashboard load time (initial) $\leq$ 5 seconds on standard broadband connection |
| | NFR-P02 | API response time (p95) $\leq$ 500 ms for standard queries |
| | NFR-P03 | Model inference time $\leq$ 200 ms per prediction |
| | NFR-P04 | Simultaneous user support: $\geq$ 50 concurrent users |
| **Availability** | NFR-A01 | System availability $\geq$ 99% during business hours (post-deployment) |
| | NFR-A02 | Scheduled daily maintenance window: 2:00 AM – 3:00 AM EET |
| **Security** | NFR-S01 | User authentication required for admin and data export functions |
| | NFR-S02 | API endpoints secured with API keys for authenticated access |
| | NFR-S03 | Data encryption at rest and in transit (TLS 1.2+) |
| | NFR-S04 | No storage of personally identifiable information (PII) or contact information |
| **Usability** | NFR-U01 | Dashboard design shall follow UI/UX best practices (consistent colors, clear typography, intuitive navigation) |
| | NFR-U02 | Supported screen resolutions: 1366x768, 1920x1080, and responsive mobile view |
| | NFR-U03 | Dashboard shall include built-in tooltips and help documentation |
| **Data Quality** | NFR-DQ01 | At least 90% data completeness for core fields (price, area, bedrooms, location) |
| | NFR-DQ02 | Clear documentation of data sources and data limitations for users |
| **Scalability** | NFR-SC01 | Data pipeline shall handle up to 100,000 new property records per month |
| | NFR-SC02 | Database shall be index-optimized for queries on location, price range, property type |
| **Maintainability** | NFR-M01 | Code shall be documented with docstrings and comments following PEP 8 |
| | NFR-M02 | README file with setup and execution instructions provided |

# 4 System Analysis and Design

## 4.1 Problem Statement and Software Architecture

### 4.1.1 Problem Statement

The Egyptian real estate market, while economically significant, remains characterized by fragmentation, information asymmetry, and limited analytical accessibility. Currently, no comprehensive, interactive, and data-driven platform exists for stakeholders (investors, analysts, policymakers, home buyers) to explore market trends, compare prices across locations, or obtain predictive price estimates. This gap results in suboptimal decision-making, inefficient capital allocation, and missed investment opportunities.

The problem addresses three core dimensions:

1. **Data Fragmentation**: Real estate information is distributed across multiple websites, each with different data schemas, update frequencies, and access methods.
2. **Analytical Inaccessibility**: Technical barriers prevent non-analysts from exploring data flexibly and generating insights independently.
3. **Predictive Capacity**: No accessible, data-driven price estimation tool exists for the Egyptian market.

### 4.1.2 Solution Approach

We propose building an automated data pipeline that collects, integrates, and analyzes real estate data from multiple Egyptian platforms, combined with official statistics. The system will provide: (a) interactive dashboard for visual exploration, (b) SQL/API access for analysts, (c) price prediction capabilities using machine learning, and (d) comprehensive market reports.

### 4.1.3 Software Architecture: Layered (MVC-based) Design

The system adopts a layered architecture inspired by the Model-View-Controller (MVC) pattern, adapted for data analysis environments. This architecture ensures separation of concerns, maintainability, and scalability.

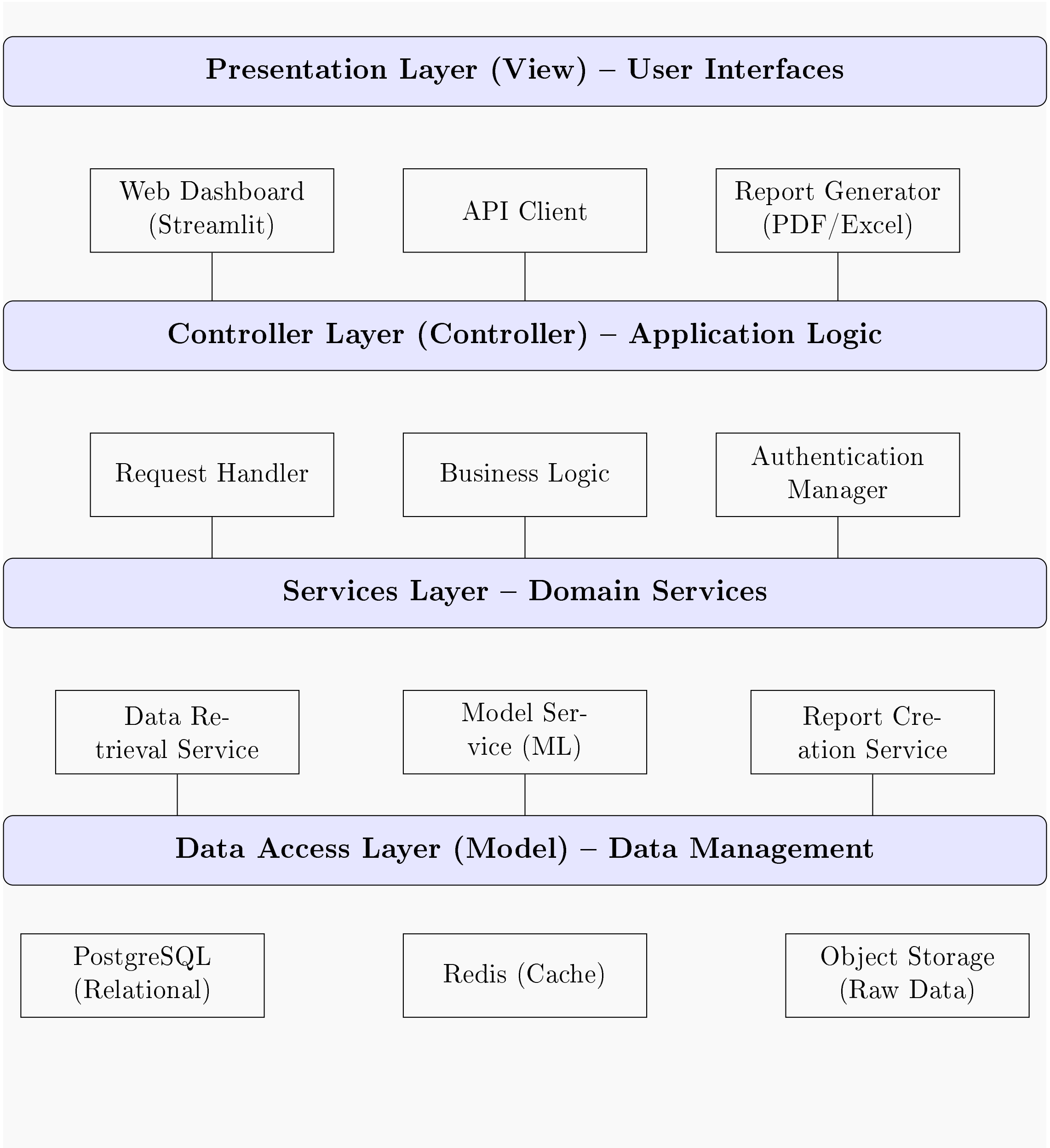



Figure 2: Layered Architecture Diagram (MVC-based Design) – Widened

## 4.2 Database Design and Data Modeling

### 4.2.1 Conceptual Data Model – Entity-Relationship (ER) Diagram

Note: Database selection is still under consideration. The ER diagram below illustrates a provisional conceptual model. It will be finalized once the specific database technology is chosen.

### 4.2.2 Logical Database Schema

The logical schema defines tables, columns, data types, and constraints:

Table 11: Table: `properties`

| **Column Name** | **Data Type** | **Constraints** | **Description** |
|---|---|---|---|
| property_id | SERIAL | PRIMARY KEY | Surrogate unique identifier |
| source_id | VARCHAR(50) | NOT NULL | Original listing ID from source website |
| source_name | VARCHAR(50) | NOT NULL | Platform name (Aqarmap/PropertyFinder) |
| title | TEXT | | Listing title text |
| property_type | VARCHAR(50) | NOT NULL | Apartment/Villa/Townhouse |
| price | DECIMAL(12,2) | NOT NULL | Price in EGP (listing price) |
| area_sqm | DECIMAL(8,2) | NOT NULL | Area in square meters |
| price_per_sqm | DECIMAL(10,2) | GENERATED ALWAYS AS (price / area_sqm) | Computed column for price per square meter |
| bedrooms | INTEGER | | Number of bedrooms (NULL for commercial) |
| bathrooms | INTEGER | | Number of bathrooms |
| floor_number | INTEGER | | Floor number (1-based; 0 = ground) |
| total_floors | INTEGER | | Total floors in building |
| furnished | BOOLEAN | DEFAULT FALSE | Furnished status |
| compound_name | VARCHAR(100) | | Name of compound/master-planned community |
| payment_method | VARCHAR(50) | | Cash/Installment/Mortgage |
| delivery_date | DATE | | Projected delivery date (new units) |
| description | TEXT | | Full listing text description |
| created_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Record creation timestamp |
| updated_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Last update timestamp |
| is_active | BOOLEAN | DEFAULT TRUE | Whether listing remains active |

Table 12: Table: `locations`

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| location_id | SERIAL | PRIMARY KEY | Surrogate unique identifier |
| city | VARCHAR(50) | NOT NULL | City name (Cairo/Alexandria/Giza) |
| district | VARCHAR(100) | NOT NULL | District/neighborhood name |
| suburb | VARCHAR(100) | | Suburb or compound area |
| latitude | DECIMAL(10,6) | | Geographic latitude (for mapping) |
| longitude | DECIMAL(10,6) | | Geographic longitude (for mapping) |
| postal_code | VARCHAR(20) | | Postal code (if available) |

Table 13: Table: `transactions`

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| trans_id | SERIAL | PRIMARY KEY | Surrogate unique identifier |
| property_id | INTEGER | FOREIGN KEY (REFERENCES properties) | |
| transaction_type | VARCHAR(20) | NOT NULL | Sale/Rent/Commercial |
| price_egp | DECIMAL(12,2) | NOT NULL | Transaction price or monthly rent |
| recorded_date | DATE | NOT NULL | Date of listing or transaction |
| currency | VARCHAR(5) | DEFAULT 'EGP' | Currency code |
| listing_url | TEXT | | Original URL of the listing |

Table 14: Table: `price_history`

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| price_id | SERIAL | PRIMARY KEY | Surrogate unique identifier |
| property_id | INTEGER | FOREIGN KEY (REFERENCES properties) | |
| recorded_date | DATE | NOT NULL | Date of price recording |
| listing_price | DECIMAL(12,2) | NOT NULL | Price at time of recording |
| price_per_sqm | DECIMAL(10,2) | | Generated from price / area |
| status | VARCHAR(20) | | Active/Sold/Withdrawn/Expired |

Table 15: Table: `scrape_log`

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| log_id | SERIAL | PRIMARY KEY | Surrogate unique identifier |
| source_name | VARCHAR(50) | NOT NULL | Source portal name |
| run_timestamp | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Timestamp of scraping run |
| records_fetched | INTEGER | NOT NULL | Count of new/updated properties |
| success_count | INTEGER | NOT NULL | Number of successful extractions |
| error_count | INTEGER | NOT NULL | Number of failed extractions |
| execution_time_seconds | INTEGER | | Duration of scrape run |
| status | VARCHAR(20) | NOT NULL | Success/Partial/Failed |
| error_log | TEXT | | Detailed error messages |

Table 16: Table: `economic_indicators`

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| econ_id | SERIAL | PRIMARY KEY | Surrogate unique identifier |
| indicator_date | DATE | NOT NULL | Date of indicator value |
| indicator_name | VARCHAR(50) | NOT NULL | Indicator type (e.g., INFLATION_RATE) |
| indicator_value | DECIMAL(10,4) | NOT NULL | Numeric value of indicator |
| unit | VARCHAR(20) | | Percentage/Index Absolute |
| source | VARCHAR(100) | | Data source (CBE/CAPMAS/World Bank) |

### 4.2.3 Data Flow Diagram (DFD) – Level 0 Context Diagram

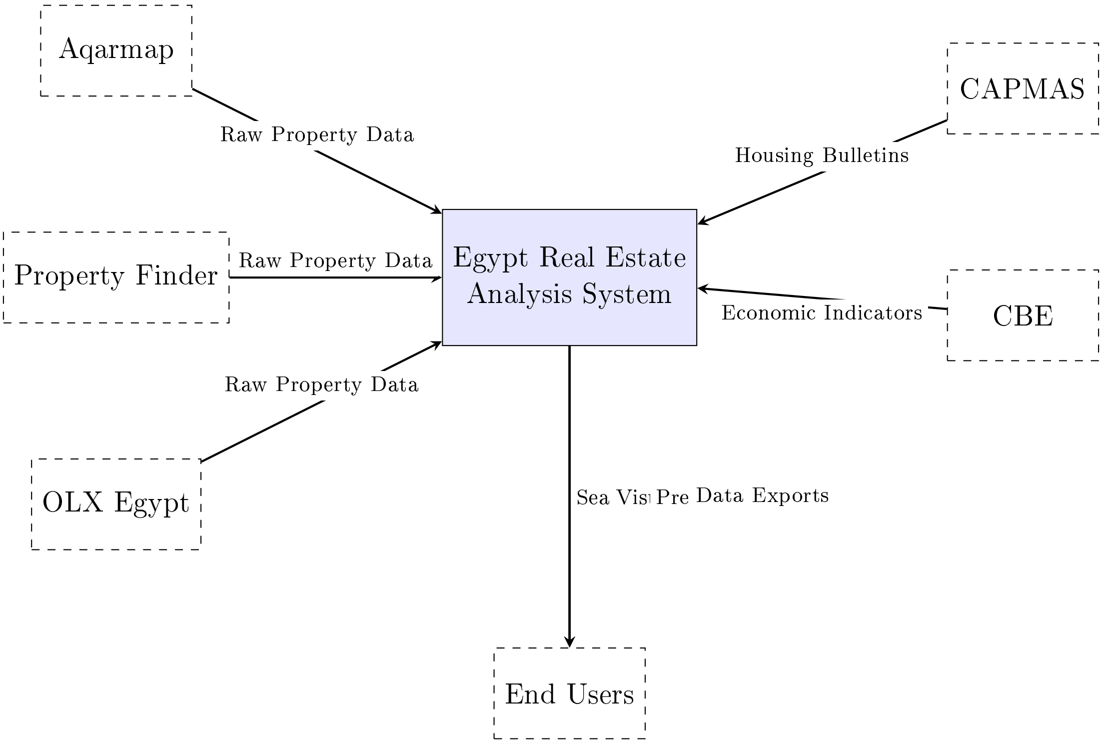



Figure 3: Context Diagram (DFD Level 0)

### 4.2.4 Wireframe – Dashboard Main View

The following table provides a high-level overview of the dashboard layout structure. All design components follow standard UI/UX analysis dashboard best practices.

Table 17: Dashboard Wireframe – Component Layout

| Section | Content | Purpose |
|---|---|---|
| Header | Project title: Egypt Real Estate Market Analysis; Date range picker; Support for light/dark mode | Branding and session control |
| Filter Panel (Left Side-bar) | Location dropdowns (City → District); Price range sliders; Property type checklist; Area range (sqm); Bedroom selector; Furnished toggle; Search/Reset buttons | Enable targeted exploration and narrowing of data view |
| Main Content Area – KPI Row (Top) | Summary metrics (Total properties, Average price, Average price per sqm, Median price, Properties added last 7 days) | At-a-glance snapshot of key metrics (best practice: 3–6 KPIs) |
| Main Content Area – Primary Chart (Center-Left) | Price distribution by location (bar or histogram); Primary chart closest to top-left as per dashboard best practices | Central data insight visualization |
| Main Content Area – Secondary Visualizations (Center-Right) | Price vs. Area scatter plot; Bedroom count distribution pie chart | Supplementary data perspectives |
| Main Content Area – Lower Section | Time-series price trend (line chart); Geographic heatmap (if coordinates available); Model performance metrics | Temporal and spatial analysis |
| Footer | Data source attribution; Last update timestamp; Generated by: `egy-real-estate-project` | Transparency and compliance |

#### 4.2.5 UI/UX Design Guidelines

The dashboard adheres to the following design principles and constraints, derived from established UX best practices for analytical dashboards:

1. **Visual Hierarchy:** Primary metrics appear largest, positioned top-left. Secondary visualizations receive visual prominence but serve supportive roles.

2. **Consistent Typography:** Sans-serif fonts (Inter or Roboto) for UI elements; monospace (Fira Code) for code/API examples. Font size hierarchy: Title (24px) → Headers (18px) → Body (14px) → Metadata (12px).

3. **Color Palette:**
   - Primary: Navy Blue (#0A2540) – Headers, primary navigation
   - Secondary: Teal (#006A71) – Action buttons, highlights
   - Accent: Coral Red (#E76F51) – Warnings, critical alerts, high-price markers
   - Neutral: Gray scale – Backgrounds, borders, secondary elements
   - Data colors: ColorBrewer qualitative palette for categorical data
4. **Responsive Layout:** Grid-based design (12-column layout) utilizing flexbox/Grid CSS; collapses to single column on mobile breakpoints.
5. **Accessibility:** WCAG 2.1 Level AA compliance: minimum contrast ratio of 4.5:1 for standard text, ARIA labels on interactive elements, keyboard navigability, screen reader support for all visualizations.
6. **Minimalist Data Presentation:** Show only necessary data points; avoid chartjunk; target 3–6 charts per screen for clarity.
7. **Loading and Error States:** Skeletons/loaders for charts; explicit error messages for failed loads; retry mechanisms.
8. **Help Integration:** Tooltips (i icons) explain metrics; link to documentation page (Markdown-based guide).

### 4.3 Missing Diagrams – Recommendations and Resources

The following standard UML and system modeling diagrams are required for complete system documentation but were not included in this LaTeX code due to length and generation complexity. They should be developed using the recommended tools below.

1. **Sequence Diagrams:**
   - *Purpose:* Show object interactions in time sequence for specific system functions (e.g., generating a price prediction).
   - *Tools:* draw.io (free, integrates with Google Drive), Lucidchart (advanced collaborative features), sequencediagram.org (simple text-based).
   - *Mermaid syntax example (renders in markdown):*

```
sequenceDiagram
    User->>Dashboard: Enter property details
    Dashboard->>API: POST /predict
    API->>Model Service: predict(features)
```

```
Model Service-->>API: price_prediction
API-->>Dashboard: response
Dashboard-->>User: display result
```

2. **Activity Diagrams:**
   - *Purpose:* Model workflows, specifically the data scraping execution pipeline (start → fetch page → parse → store → check next page → end).
   - *Tools:* draw.io, PlantUML (text-based), Visual Paradigm (academic license available).
   - *Key visual elements:* Start/end nodes, activities (rounded rectangles), decision diamonds, parallel fork/join bars.
3. **State Diagrams:**
   - *Purpose:* Show possible states of a system object (e.g., "Model Training → Training → Validated → Deployed → Archived" or "Data Pipeline → Scheduled → Running → Completed").
   - *Tools:* yEd Live (free desktop), draw.io, UMLet (lightweight).
   - *Alternative approach:* Represent states as nodes with transitions as labeled arrows.
4. **Class Diagram:**
   - *Purpose:* Show system's object-oriented structure: classes, attributes, methods, and relationships (inheritance, association, composition). For the Python implementation, the central class is `DataProcessor` (methods: `clean_data()`, `extract_features()`).
   - *Tools:* Built into most IDEs (PyCharm, VS Code), Pylint extensions, PlantUML, StarUML (free, diagrams-focused).
5. **Deployment Diagram:**
   - *Purpose:* Map hardware and software: physical server nodes, Docker containers, cloud services (AWS EC2 + RDS), and connections.
   - *Tools:* Cloudcraft (AWS-specific), draw.io (with AWS/GCP icons), Lucidchart (cloud architecture templates).
6. **Component Diagram:**

- *Purpose:* Show high-level software components and their dependencies: Database, ETL Pipeline, API Gateway, Dashboard Frontend.
- *Tools:* Structurizr (C4 model, code-based diagrams), draw.io, PlantUML.
- *Alternative:* Represent components as rectangles with connectors, using the layered architecture diagram (included above) as a foundation.